What Makes The Cut? Interpretable Shot Selection and Clustering of Movie Trailers

Project Update | Team 13: Cedric Lam, Estee Ng | github.com/lvtwoodpecker/trailercut-vizml

Progress

- **Data.** 293 MovieNet films with keyframes, synopsis-to-shot alignment and a trailer, all 293 with a working trailer: 49 dead links were replaced by YouTube search, accepted only if at least 25% of the trailer's shots align to that film. The repo stores YouTube IDs, logs, scripts and features, not videos.
- **Picture alignment.** CLIP embeddings of 3 keyframes per shot; each trailer shot is matched to its best film shot or to "no match". The cutoff comes from 710 hand-labeled trailer shots: precision 0.85, recall 0.91 (the top match is right 94% of the time when a source exists). Sequence smoothing and a copy-detection re-ranker were tested against the labels and dropped.
- **Sound alignment.** Whisper transcripts of every trailer, matched to the film's subtitles: 29% of speech segments are found in the film. Only about half of those lines play over their own scene (51%); the rest are laid over footage from elsewhere.
- **RQ1.** 464K film shots, 3.8% used in a trailer. LightGBM with folds grouped by film picks the trailer's exact shots 3.2× as often as chance. Ablations: CLIP appearance adds the most, then editing (shot length), then story position. Trailers favor the start of the film (1.5× for the first 5%) and avoid the late film (0.6× around 90%).
- **Clustering.** Stable (bootstrap ARI 0.79 to 0.94) but weakly separated (silhouette 0.11 to 0.31); pacing clusters follow decade, dialogue clusters follow genre.
- **Views.** D3 Trailer View: the trailer plays with its timeline, each shot linked to its source on a "movie barcode" of the film, sound linked separately, plus an RQ1 page of what gets picked across all films.

Response to Feedback

- **Label quality.** Every trailer shot gets an explicit no-match option. The starting threshold comes from scoring each shot against a *different* film; blank shots are masked on both sides and letterbox bars are cropped per trailer. Since that null cannot see "right scene, wrong shot" errors, we hand-labeled 10 decade-stratified trailers (710 usable shots) against each shot's top-5 candidates and picked the threshold from that precision/recall curve; a second labeler on 2 of them will give Cohen's κ. The augmentation target of 0.9 is dropped.
- **Attributions.** Visual features are readable by construction: zero-shot shot scale (5 MovieNet classes) and 20 named content concepts, plus story position, dialogue sentiment and editing features. SHAP is shown grouped (position, dialogue, scale, content, editing, "other appearance (CLIP)"), with the CLIP block never split into dimensions. We report readable-only vs. readable+CLIP AUPRC so the gap is visible. Shot length is a known confound (longer shots offer more footage) and is controlled for.
- **Scope.** RQ1 and the Trailer View are the core. Clustering becomes one supporting Overview; the spoiler stretch goals are dropped, with final-act share kept as a labeled proxy.
- **Clustering evaluation.** Bootstrap ARI for stability and NMI with genre/decade against a permutation null. Since clusters partly follow decade, we will regress genre and decade out and re-cluster the residuals; if structure stays weak, we report continuous axes instead of trailer "types".
- **Trailer data.** IDs, availability log and features are committed; everything else is rebuilt by one script.

Scope and Plan

Nov 3 to 17	Second labeler for agreement; sound/picture features in the clustering; residual clustering; cluster Overview view; custom player with the timeline as scrubber.
Nov 17 to Dec 1	Overview linked to the Trailer View; case studies (model misses, sound/picture mismatches); ViT-H/14 rerun on the cluster.
Dec 1 to 15	Ablations, final evaluation, report and presentation.

Who Does What

Cedric	Data pipeline (MovieNet download, trailer scraping and recovery, shot detection, CLIP, alignment, cluster jobs), RQ1 trailerness model and SHAP, Trailer View.
Estee	RQ2 clustering (feature selection, k-means/HDBSCAN, stability and genre/era baselines, residual clustering), cluster Overview in D3 and its link to the Trailer View, case study selection.
Both	Hand labeling (shared trailers for agreement), report and presentation.